



OPEN

DATA DESCRIPTOR

MediQAI: A French Medical Question Answering Dataset for Knowledge and Reasoning Evaluation

Adrien Bazoge^{1,2}

This work introduces MediQAI, a French medical question answering dataset designed to evaluate the capabilities of language models in factual medical recall and reasoning over real-world clinical scenarios. MediQAI contains 32,603 questions sourced from French medical examinations across 41 medical subjects. The dataset includes three tasks: (i) Multiple-Choice Question with Unique answer, (ii) Multiple-Choice Question with Multiple answer, and (iii) Open-Ended Question with Short-Answer. Each question is labeled as *Understanding* or *Reasoning*, enabling a detailed analysis of models' cognitive capabilities. We validate the MediQAI dataset through extensive evaluation with 14 large language models, including recent reasoning-augmented models, and observe a significant performance gap between factual recall and reasoning tasks. Our evaluation provides a comprehensive benchmark for assessing language models' performance on French medical question answering, addressing a crucial gap in multilingual resources for the medical domain.

Background & Summary

Medical licensing examinations, originally designed to assess students' knowledge and reasoning, are increasingly repurposed as benchmarks for evaluating large language models (LLMs) medical capabilities¹. Benchmarks of question-answering tasks predominantly rely on multiple-choice questions (MCQs) with a single correct answer^{2,3}. This format is widely used due to the availability of automatic evaluation metrics that provide consistent and objective assessment of LLMs at scale.

While MCQ-based datasets and metrics provide a valuable initial insight into LLM performance, they are often limited in several key aspects: the number of examples, the diversity of medical subjects covered, their representation of real-world clinical scenarios⁴, and the range of languages represented. Indeed, most of existing benchmarks are heavily centered around English, which restricts their applicability to multilingual or non-English contexts¹. Furthermore, medical benchmarks inherently reflect cultural, educational, and regulatory contexts in which they are developed. The format of questions and answers mirrors how medicine is taught and assessed in their respective countries, which differs in structure, emphasis, and evaluative expectations across regions. In addition, treatment guidelines, clinical protocols, and legal standards are often country-specific, meaning that identical questions translated across languages can pose entirely different challenges. In this regard, French medical examinations illustrate how linguistic and pedagogical specificities can further shape the difficulty of medical question answering. French medical examination questions are typically characterized by long and syntactically complex sentences, frequent use of embedded clauses, and heavy reliance on negation and exception-based reasoning (e.g. "All of the following statements are true, except one. Which one?"). In addition, French clinical discourse often encodes temporal and causal relationships implicitly rather than through explicit markers, increasing the difficulty of clinical reasoning required to interpret the questions.

Multiple-choice question answering with a unique correct answer (MCQU) is a well-established task, frequently used to benchmark language models. In English, multiple high-quality datasets exist, including HEAD-QA⁵, MedQA⁶, MedMCQA⁷, PubMedQA⁸, MMLU (Medical)^{2,3}. Similar efforts have emerged in other

¹Nantes Université, CHU Nantes, Pôle Hospitalo-Universitaire 11: Santé Publique, Clinique des données, INSERM, CIC 1413, F-44000, Nantes, France. ²Nantes Université, Ecole Centrale Nantes, CNRS, LS2N, UMR 6004, F-44000, Nantes, France. e-mail: adrien.bazoge@univ-nantes.fr

	Train	Validation	Test
MCQU	10,113	2,561	4,343
MCQM	5,767	1,466	3,384
OEQ	—	—	4,969

Table 1. MediQAI dataset distribution.

languages, with MCQ datasets in Chinese⁹, Polish¹⁰ and Spanish¹¹. However, French remains underrepresented. FrenchMedMCQA¹² is a dataset containing 3,105 multiple-choice questions, with both unique and multiple answers, but is limited to pharmacy topics. MedExpQA¹¹ includes a French subset that is translated from Spanish and contains 622 examples. FrBMedQA¹³ is a dataset of 41,000 passage-questions automatically built from French biomedical Wikipedia articles. Open-ended question answering (OEQ) in medicine is even less common, even in English, due to the need for manual evaluation. For example, MEDQA-OPEN¹⁴ reformulates MedQA into an open-ended format, but for French, the only available OEQ dataset is MedFrenchmark¹⁵, which contains only 114 examples.

Recent efforts have focused on increasing diversity in question difficulty and covering a wider variety of medical subjects¹⁶, yet these benchmarks remain predominantly limited to English-language and rely on MCQs with a single correct answer. This limitation is particularly problematic, as several studies have demonstrated significant performance disparities between languages, with LLMs performing considerably better in English compared to less-resourced languages^{11,17,18}. Therefore, it is crucial to develop more inclusive benchmarks that cover a broader range of languages and are more reflective of real-world clinical scenarios, ensuring a fair and comprehensive evaluation of LLMs in the medical domain.

In this work, we present MediQAI, a medical question answering dataset for French. This dataset contains questions sourced from French medical licensing examinations. These are manually created by academic and hospital faculty members to reflect real-world clinical scenarios and cover a broad range of medical subjects. This paper makes the following contributions:

1. We introduce MediQAI, a French medical question answering (QA) dataset that includes three tasks: (i) Multiple-Choice Question with Unique Answer (MCQU), (ii) Multiple-Choice Question with Multiple Answers (MCQM) and (iii) Open-Ended Question with Short-Answer (OEQ).
2. MediQAI covers a total of 41 medical subjects and each question is categorized as either *Understanding* or *Reasoning*, enabling detailed analysis of LLMs' capabilities across different cognitive tasks.
3. We present an extensive evaluation of 14 large language models (LLMs) on MediQAI, including latest reasoning-based models, providing a comprehensive benchmark for assessing their performance over real-world clinical scenarios. We compare different groups of models, focusing on the performance gap between vanilla models (non-reasoning) and their reasoning-enhanced counterparts.

Methods

Data collection. For the construction of this dataset, the raw data was collected from publicly available websites and forums where professors and students share examination questions intended for training purposes in preparation for the national French medical examination, such as ECN exams. The National Classifying Tests (*Épreuves Classantes Nationales - ECN*) are the theoretical exams conducted during the sixth year of medical studies in France. These exams determine the ranking of medical students, which in turn allows them to select their university hospital for residency, as well as their specialization track and the services where they will complete six-month clinical internships. The ECN also serves as a comprehensive evaluation of the students' medical knowledge and clinical reasoning, crucial for their future roles as medical practitioners. The exam consists of the following components: (i) clinical scenario-based questions, (ii) isolated knowledge-based questions, and (iii) critical article analysis questions. The question formats include multiple-choice questions (MCQs) with five options (either a single correct answer or multiple correct answers) and open-ended short-answer questions. Each year, examination questions and answer are manually created and verified by a scientific advisory board composed of tenured academic and hospital faculty members.

In line with this structure, we organized the dataset into multiple subsets corresponding to the ECN's questions formats. The multiple-choice questions (MCQU and MCQM) were automatically extracted from the qcm-lab website (<https://qcm-lab.net/revision.php>) in March 2024. Each instance in these subsets contains a unique ID, an optional clinical scenario, a question, five candidates, the associated medical subject and the correct answer.

For the open-ended questions with short-answer (OEQ subset), the raw data was collected from multiple sources as HTML and PDF files. HTML files were well-structured enough to automatically extract QA instances using regular expressions. For the remaining PDF files, their structure was not homogeneous and could not be parsed automatically, and each instance was then extracted and curated manually.

Data filtering. To ensure that the collected instances were homogeneous and that the questions were answerable, we applied several filters and preprocessing steps.

For the MCQU and MCQM subsets, 547 questions with missing correct answers or candidate options were removed. Since these subsets contained a large number of questions, we used three models (Llama-3.1-8B-Instruct¹⁹, Qwen2.5-7B-Instruct²⁰ and Mistral-7B-Instruct-v0.3²¹) to vote on and filter questions to only

Medical Subjects Annotation Prompt

You are an experienced medical doctor and independent practitioner. Your task will be to label a clinical scenario according to the medical subject it corresponds to.

You will be given a list of medical subjects, followed by a clinical scenario. Please determine which subject the clinical scenario best pertains to. If the clinical scenario is related to multiple subjects, only select the most relevant one.

Directly output the name of the final subject you selected from the list of available subjects.

****Subjects:****

Cardiologie et Pathologie Vasculaire, Hépatogastro-Entérologie, Pneumologie, Néphro-Urologie, Psychiatrie, Hématologie, Endocrinologie-Métabolisme, Gynécologie-Obstétrique, Rhumatologie, Neurologie, Maladies Infectieuses, Dermatologie, Pédiatrie, Oto-Rhino-Laryngologie, Ophtalmologie, Immunologie, Orthopédie, Pharmacologie, Médecine Légale et Toxicologie, Anatomie pathologique, Biochimie, Epidémiologie, Génétique, Médecine du Travail, Microbiologie, Parasitologie, Rééducation, Physiologie, Histologie, Radiologie, Cytologie, Embryologie, Anatomie, Urgences

****Clinical scenario:****

{clinical_scenario}

****Output:****

Fig. 1 Prompt for Medical Subjects Annotation in the MediQAI-OEQ subset. The list of medical subjects in the prompt was made from the list of medical subjects from MCQU and MCQM subsets.

Understanding/Reasoning Annotation Prompt

You are an experienced medical doctor and independent practitioner. Your task will be to determine whether a medical question primarily challenges the answerer's medical knowledge understanding or medical reasoning skills.

You will be given the question. Please determine whether the question primarily challenges the answerer's medical knowledge understanding or medical reasoning ability.

Reasoning : complicated, reasoning-heavy questions.

Understanding : little to no reasoning and instead assess skills such as medical knowledge.

Directly output either "Understanding" Or "Reasoning" as your answer without any additional information or explanations.

****Clinical scenario:****

{clinical_scenario}

****Question:****

{question}

****Output:****

Fig. 2 Prompt for labeling questions as *Understanding* or *Reasoning*.

keep challenging questions in the test sets. If any of the models answer a question correctly, the question is deemed too simple and is removed from the test sets. All filtered-out questions were then randomly split into training (80%) and validation (20%) sets for both MCQU and MCQM subsets. The dataset splits for all tasks are presented in Table 1.

For the OEQ subset, questions with clinical scenario containing images or tables were removed. Points awarded for each response element, sometimes embedded in the response text, were removed. Duplicates questions were identified by calculating cosine similarity on TF-IDF²² vectorized representations of both questions and answers. All QA pairs with a similarity score greater than 0.70 were manually reviewed, and duplicates were removed. To retain only short-answer questions, we tokenized each answer using a French medical tokenizer from DrBERT model²³ and excluded instances where the length of the answer exceeded 200 tokens. All these filtering steps removed a total of 283 questions from the OEQ subset.

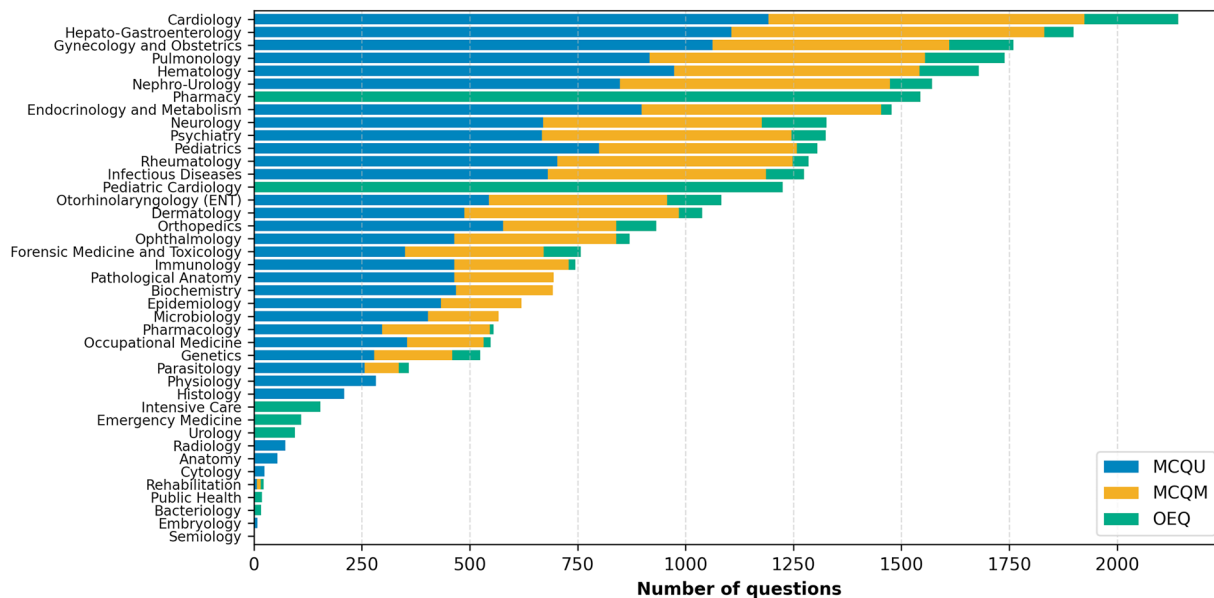


Fig. 3 Distribution of medical subjects across MediQAI dataset.

MCQU Example

Question: L'hypertension artérielle maligne se caractérise par les éléments suivants, sauf un, lequel ?

- (A) Une hypertension artérielle surtout diastolique
- (B) Un stade II au fond d'oeil
- (C) Des manifestations d'insuffisance ventriculaire gauche
- (D) Un retentissement neuro-sensoriel majeur
- (E) Une insuffisance rénale

Réponse: B

Question: Malignant arterial hypertension is characterized by the following elements, except one. Which one?

- (A) Predominantly diastolic arterial hypertension
- (B) Stage II changes on fundoscopic (eye) examination
- (C) Signs of left ventricular failure
- (D) Major neuro-sensory impairment
- (E) Renal insufficiency

Answer: B

Fig. 4 Example of Multiple-Choice Question with unique answer (MCQU) from MediQAI dataset with English translation.

	MCQU			MCQM			OEQ		
	Understanding	Reasoning	Total	Understanding	Reasoning	Total	Understanding	Reasoning	Total
Total number of questions	11,336	5,681	17,017	7,742	2,875	10,617	1,842	3,125	4,969
# Isolated questions	9,126	961	10,087	6,200	343	6,543	836	179	1,015
# In-context questions	2,210	4,720	6,930	1,542	2,532	4,074	1,006	2,946	3,954
Avg. Question length	18.95	21.57	19.82	13.20	16.12	13.99	16.79	20.95	19.40
Avg. clinical scenarios length	83.50	107.67	99.97	94.87	114.77	107.24	109.71	141.28	132.19
Avg. answer length	—	—	—	—	—	—	25.26	40.24	34.68

Table 2. Characteristics of the MediQAI dataset. In-context questions refer to questions including a clinical scenario. Lengths are measured in words.

MCQM Example

Cas clinique: Un sujet de sexe masculin, âgé de 55 ans, ouvrier spécialisé en mécanique, se plaint d'un oeil droit douloureux et rouge depuis 48 h. A l'examen, on retrouve au niveau de l'oeil droit des signes d'iridocyclite. La vision est égale à 8/10 oeil droit, 10/10 oeil gauche après correction d'une hypermétropie de + 1,5 dioptrie. Le tonus oculaire est égal à 28 mmHg oeil droit et à 16 mmHg oeil gauche. L'examen gonioscopique met en évidence un angle irido-cornéen ouvert à droite comme à gauche. Le fond d'oeil est normal des deux côtés. Dans les antécédents du patient, on retrouve une symptomatologie évoquant une spondylarthrite ankylosante et la notion de projections métalliques au cours du travail

Question: Le traitement de l'iridocyclite comportant l'instillation répétée de collyre à base de cortisone, quelle(s) complication(s) peut-on voir survenir ?

- (A) Cataracte
- (B) Hypertonie
- (C) Dystrophie cornéenne
- (D) Sclérose des canalicules lacrymaux
- (E) Conjonctivite papillaire

Réponse: A,B

Clinical case: A 55-year-old male, a specialized mechanical worker, complains of a painful and red right eye for 48 hours. On examination, signs of iridocyclitis are found in the right eye. Visual acuity is 8/10 in the right eye and 10/10 in the left eye after correction of +1.5 diopters of hypermetropia. Intraocular pressure is 28 mmHg in the right eye and 16 mmHg in the left eye. Gonioscopic examination shows an open iridocorneal angle on both the right and left sides. Fundus examination is normal in both eyes. In the patient's medical history, there are symptoms suggesting ankylosing spondylitis and a history of metallic projections during work.

Question: The treatment of iridocyclitis includes repeated instillation of cortisone-based eye drops. Which complication(s) may occur?

- (A) Cataract
- (B) Ocular hypertension
- (C) Corneal dystrophy
- (D) Sclerosis of the lacrimal canaliculi
- (E) Papillary conjunctivitis

Answer: A,B

Fig. 5 Example of Multiple-Choice Question with multiple answers (MCQM) from MediQAI dataset with English translation.

Data quality assessment. The quality of the remaining questions after automatic filtering was assessed by conducting a human evaluation with a medical expert. We selected a stratified sample of 150 questions (50 from each test set) that were reviewed. The medical expert judged 124/150 questions to be correct. Nine questions were labeled as incorrect, identifying instances where the answer did not align with established medical knowledge. Seven questions were marked as outdated, meaning that while the answer was consistent with medical guidelines at the time of data creation, current recommendations have since changed. Four questions were found to lack sufficient clinical context to support a definitive answer, due to the use of isolated questions from clinical case that are sometimes multi-step. Finally, six questions were categorized as out of the expert's medical domain and could not be reliably assessed.

Medical subject annotation. For MCQU and MCQM subsets, the medical subject for each question was directly available in the collected data sources. However, for the OEQ subset, the medical subjects were not consistently present across all data sources. To address this, we instructed GPT-4o (gpt-4o-2024-08-06)²⁴ to automatically assign a medical subject to questions when it was missing. The prompt used for this annotation is provided in Fig. 1.

Understanding and reasoning questions. To assess the capacity of LLMs to handle complex clinical reasoning tasks beyond simple recall of medical knowledge, we implemented an automatic question categorization

OEQ Example

Cas clinique: E.P., âgée de 12 ans est née en Grèce et séjourne en France depuis 1 an. Elle se plaint d'une gêne de l'hypochondre gauche lorsqu'elle fait de la bicyclette. L'examen clinique montre une importante splénomégalie. L'imagerie médicale révèle dans la rate une masse kystique liquidienne et au niveau du foie deux autres kystes de même aspect : Le diagnostic d'hydatidose est évoqué.

Question: Quels sont les risques majeurs de cette complication ?

Réponse: Choc anaphylactique (mort) et risque d'échinococcose secondaire

Clinical case: E.P., a 12-year-old girl, was born in Greece and has been living in France for one year. She complains of discomfort in the left hypochondrium when she rides a bicycle. The clinical examination shows significant splenomegaly. Medical imaging reveals a cystic fluid-filled mass in the spleen and two other cysts of the same appearance in the liver. A diagnosis of hydatidosis is considered.

Question: What are the major risks of this complication?

Answer: Anaphylactic shock (death) and risk of secondary echinococcosis

Fig. 6 Example of Open-Ended Question with a short answer (OEQ) from MediQAI dataset with English translation.

MCQU Prompt

You are an experienced medical doctor and independent practitioner. Your task is to answer the following medical multiple-choice question. There is only one correct choice. The last line of your response should be of the following format: 'Answer: \$LETTER' (without quotes) where LETTER is one of ABCDE. Think step by step before answering.

****Clinical scenario:****

{clinical_scenario}

****Question:****

{question}

(A) {option_A}

(B) {option_B}

(C) {option_C}

(D) {option_D}

(E) {option_E}

Fig. 7 Prompt for zero-shot evaluation of LLMs on the MCQU subset. The clinical scenario is optional in the prompt.

approach. Specifically, we categorized each question into one of two types: *Understanding* or *Reasoning*. This categorization was performed using GPT-4o, following the strategy outlined by Zuo *et al.*¹⁶. The details of the prompt used for this process are provided in Fig. 2.

The quality of this automatic categorization was manually assessed by reviewing 10 randomly selected questions for each medical subject (5 labeled as *Understanding*, and 5 as *Reasoning*) from the test set of each task. In total, 858 questions were reviewed. Among these, 72 questions were explicitly mislabeled, resulting in an error rate of 8.4%.

Data Records

MediQAI²⁵ is available on Hugging Face (<https://huggingface.co/datasets/ANR-MALADES/MediQAI>) and Zenodo (<https://doi.org/10.5281/zenodo.18220039>) under CC-BY-4.0 license. It consists of questions sourced from French medical examinations. MediQAI is designed to evaluate medical knowledge and reasoning on both isolated and in-context questions reflecting real-world clinical scenarios. The dataset includes three subsets, corresponding to distinct question answering tasks:

- (1) Multiple-Choice Question with unique answer (MCQU). This task can be formulated as $X = \{C, Q, (O_1, \dots, O_5), A\}$ where C is an optional clinical scenario, Q is the question, $(O_1, \dots, O_5)$ are five candidate options and A is the correct answer. For a given triplet $\{C, Q, (O_1, \dots, O_5)\}$, the correct answer A is a single option O_i from $(O_1, \dots, O_5)$. This task is similar to most existing MCQA datasets.

MCQM Prompt

You are an experienced medical doctor and independent practitioner. Your task is to answer the following medical multiple-choice question. Multiple selections are required; single-choice answers are not accepted. The last line of your response should be of the following format: 'Answer: \$LETTERS' (without quotes) where LETTERS are multiple letters of ABCDE, separated by commas (e.g., A,B,C). Think step by step before answering.

****Clinical scenario:****

{clinical_scenario}

****Question:****

{question}

(A) {option_A}

(B) {option_B}

(C) {option_C}

(D) {option_D}

(E) {option_E}

Fig. 8 Prompt for zero-shot evaluation of LLMs on the MCQM subset. The clinical scenario is optional in the prompt.

OEQ Prompt

You are an experienced medical doctor and independent practitioner. Your task is to answer the following medical question in French by providing a well-structured and concise response. The last line of your response should be of the following format: 'Answer: \$TEXT' (without quotes) where TEXT is your final answer in French. Think step by step before answering.

****Clinical scenario:****

{clinical_scenario}

****Question:****

{question}

Fig. 9 Prompt for zero-shot evaluation of LLMs on the OEQ subset. The clinical scenario is optional in the prompt.

- (2) Multiple-Choice Question with multiple answers (MCQM). This task follows a similar formulation as MCQU: $X = \{C, Q, (O_1, \dots, O_5), A\}$. However, in MCQM, the correct answer A is a subset of candidate options with $|A| \geq 2$. The answer includes multiple correct options among $(O_1, \dots, O_5)$.
- (3) Open-Ended Question with a short answer (OEQ). The OEQ task can be formulated as $X = \{C, Q, A\}$ where C is an optional clinical scenario, Q is the question and A is a short, free-text answer. The answer length is lower than 200 tokens.

MediQAI contains a total of 32,603 questions, of which 17,017 are MCQU, 10,617 are MCQM and 4,969 are OEQ. These questions span 41 medical subjects, as displayed in Fig. 3, and are categorized as *Understanding* or *Reasoning*, offering a diverse and reliable benchmark for medical question answering tasks in French. Table 2 summarizes the main characteristics of the dataset, and Figs. 4–6 illustrates examples for each of the three tasks.

Technical Validation

Experimental setup. *Models.* We evaluate several leading LLMs on MediQAI, covering both proprietary and open-source models, including vanilla models and recent reasoning-based models. This selection ensures diversity across training paradigms, architectural families, and intended use cases, allowing us to assess how reasoning specialization and medical adaptation influence performance.

Vanilla large language models. GPT-4o (gpt-4o-2024-08-06)²⁴, DeepSeek-V3²⁶, Qwen2.5-72B-Instruct²⁰, Llama-3.3-70B-Instruct¹⁹, Llama-3-UltraMedical 70B and 8B²⁷ and BioMistral-7B²⁸.

Reasoning large language models. o3 (o3-2025-04-16)²⁹, DeepSeek-R1³⁰, DeepSeek-R1-Distill-Llama-70B³⁰, HuatuoGPT-o1-8B³¹, FineMedLM-o1-8B³², DeepSeek-R1-Distill-Llama-8B³⁰, DeepSeek-R1-Distill-Qwen2.5-7B³⁰.

Model	Accuracy		
	Understanding	Reasoning	Average
Random baseline	20.20	18.71	19.02
<i>Reasoning LLMs</i>			
o3	74.76	70.63	73.15
DeepSeek-R1	<u>69.07</u>	<u>63.82</u>	<u>67.03</u>
DeepSeek-R1-Distill-Llama-70B	46.04	49.2	47.27
HuatuoGPT-o1-8B	24.4	22.62	23.49
FineMedLM-o1 (8B)	3.99	3.55	3.82
DeepSeek-R1-Distill-Llama-8B	9.31	6.63	8.27
DeepSeek-R1-Distill-Qwen2.5-7B	14.17	10.24	12.64
<i>Vanilla LLMs</i>			
GPT-4o	65.0	54.59	60.95
DeepSeek-V3	66.24	58.73	63.32
Qwen2.5-72B-Instruct	48.53	41.86	45.94
Llama3.3-70B-Instruct	46.57	38.31	43.36
Llama-3-70B-UltraMedical	41.52	33.27	38.31
BioMistral-7B	11.72	12.97	12.20
Llama-3.1-8B-UltraMedical	14.66	10.72	13.12
BioMistral-7B-SFT	27.81	27.89	27.84

Table 3. Performance of LLMs on the MediQAI-MCQU subset. The scores, obtained in zero-shot, are measured with Accuracy.

Supervised fine-tuning. In addition to all evaluated models, we conducted supervised fine-tuning (SFT) on BioMistral-7B to assess the learnability and utility of the MediQAI dataset. The BioMistral-7B-SFT model was trained for two epochs using the combined training sets of all tasks (MCQU, MCQM, and OEQ). Since the OEQ subset lacks a dedicated training set, we converted questions from MCQU and MCQM training sets into OEQ format to enable unified training. We performed full fine-tuning of the model with a learning rate of 2×10^{-5} .

Evaluation framework. All models across all tasks were evaluated in a zero-shot prompting setup, using greedy decoding for output generation when available to ensure result stability. For reasoning models that require specific evaluation settings, we followed the recommended instructions provided for each. To reduce inference time, open-source models were limited in their output length: up to 2,048 tokens for vanilla models and up to 8,192 tokens for reasoning-based models. For API-based models (o3, GPT-4o, DeepSeek-V3 and DeepSeek-R1), we followed the recommended prompting guidelines, removing the system prompt while keeping all other parameters at their default settings.

The evaluation prompts and scripts used to extract responses from the generated text were inspired by the format of the *simple-evals* framework (<https://github.com/openai/simple-evals>). The specific prompts used for MCQU, MCQM and OEQ tasks are provided in Figs. 7–9, respectively.

Random baselines. To contextualize model performance, we included simple random baselines tailored to the structure of each task. For MCQU, we randomly selected an answer from the five available options (A to E). For MCQM, we randomly sampled one combination among all possible subsets of size two to five using the five available options (A to E). For the OEQ task, we used as answer the first L tokens of the question itself, with L set to the average answer length observed in the test set.

Evaluation metrics. *Multiple-choice question with unique answer (MCQU).* For the evaluation on the MCQU subset, we used Accuracy, similarly to other single-answer multiple-choice tasks such as MMLU².

Multiple-choice question with multiple answers (MCQM). We used Exact Match Ratio (EMR) and Hamming score to evaluate multiple-choice questions with multiple answers, following previous work on this task¹². The two metrics are defined as follows:

$$\text{Exact Match Ratio(EMR)} = \frac{1}{N} \sum_{i=1}^N [\hat{y}_i = y_i]$$

where N denotes the number of questions, $\hat{y}_i$ is the set of predicted answers for the i^{th} question, y_i is the set of correct answers for the i^{th} question, and $[x]$ is an indicator function that returns 1 if x is true and 0 otherwise.

$$\text{Hamming Score} = \frac{1}{N} \sum_{i=1}^N \frac{|y_i \cap \hat{y}_i|}{|y_i \cup \hat{y}_i|}$$

Model	EMR			Hamming		
	Understanding	Reasoning	Avg.	Understanding	Reasoning	Avg.
Random baseline	4.76	3.8	4.46	41.64	39.21	40.89
<i>Reasoning LLMs</i>						
o3	56.87	51.04	55.05	80.88	77.08	79.7
DeepSeek-R1	<u>51.12</u>	<u>43.93</u>	<u>48.88</u>	<u>79.21</u>	<u>73.83</u>	<u>77.54</u>
DeepSeek-R1-Distill-Llama-70B	19.91	22.68	20.77	35.55	43.62	38.07
HuatuoGPT-o1-8B	8.28	5.22	7.33	47.35	41.42	45.5
FineMedLM-o1	1.46	0.19	1.06	18.76	6.93	15.07
DeepSeek-R1-Distill-Llama-8B	2.1	2.56	2.25	10.37	16.06	12.15
DeepSeek-R1-Distill-Qwen2.5-7B	2.27	2.28	2.28	19.03	19.48	19.17
<i>Vanilla LLMs</i>						
GPT-4o	46.48	37.37	43.65	76.03	69.64	74.04
DeepSeek-V3	49.18	39.37	46.13	78.45	72.02	76.45
Qwen2.5-72B-Instruct	31.8	26.47	30.14	67.38	61.5	65.55
Llama3.3-70B-Instruct	21.72	11.29	18.47	62.94	54.41	60.29
Llama-3-70B-UltraMedical	22.40	12.71	19.39	62.38	53.36	59.57
BioMistral-7B	0.82	1.33	0.98	5.33	12.41	7.54
Llama-3.1-8B-UltraMedical	5.11	4.55	4.93	44.53	40.39	43.24
BioMistral-7B-SFT	3.21	3.81	3.38	24.07	22.75	23.66

Table 4. Performance of LLMs on the MediQAI-MCQM subset. The scores, obtained in zero-shot, are measured in terms of Exact Match Ratio (EMR) and Hamming score.

where N denotes the number of questions, γ_i is the set of correct answers for the i^{th} question, $\hat{\gamma}_i$ is the set of predicted answers for the i^{th} question, $|\gamma_i \cap \hat{\gamma}_i|$ is the intersection size between the correct and predicted answers, and $|\gamma_i \cup \hat{\gamma}_i|$ is the size of the union of correct and predicted answers.

Open-ended question with short-answer (OEQ). To evaluate the free-text responses in the OEQ subset, we opted for a combination of lexical and contextual embedding-based metrics that align with human judgments on clinical texts³³. These metrics are: ROUGE-1³⁴, BLEU-4³⁵, and BERTScore (roberta-large-mnli)³⁶.

Given the inherent complexity of evaluating open-ended question answering task, where clinically acceptable responses can differ significantly in phrasing, we supplemented traditional metrics with an automatic evaluation using a LLM-as-Judge approach³⁷. This strategy consists of comparing model-generated responses with expert-provided references, allowing for more nuanced assessment beyond surface-level lexical and semantic similarity. To minimize bias, we adopted Gemini-2.0-Flash³⁸ as the judging model, as it was not part of the evaluated model set and does not belong to the same model family as any evaluated model.

The model was prompted to assign a score from 1 to 10 for each (question, model answer, expert answer) triplet, as described in the LLM-as-Judge prompt in Fig. 10. A score of 0 was assigned to cases where the evaluated model either failed to produce a final answer or generated a response that did not conform to the expected format and was therefore unparseable. Final scores were averaged across all examples and scaled to a 0–100 range for reporting.

Experimental results. Multiple-choice question with unique answer. Table 3 shows the performance of all evaluated LLMs on the MCQU subset of the MediQAI dataset. We observe that o3 achieves the highest performance on both *Understanding* and *Reasoning* questions with 73.15% accuracy. Among open-source models, DeepSeek-R1 and DeepSeek-V3 perform well on this subset with 67.03% and 63.32% accuracy, even surpassing some commercial models such as GPT-4o (60.95%). In contrast, models like DeepSeek-R1-Distill-Llama-70B, Llama-3.3-70B and Qwen2.5-72B demonstrate lower performance, correctly answering half of the questions in the test set. For smaller open-source models, HuatuoGPT-o1-8B is the only model that exceeds the random-choice baseline of 19.02% accuracy compared to others in the same size category, achieving 23.49%. Furthermore, BioMistral-7B-SFT, fine-tuned on the MediQAI training sets, shows substantial performance gains of 15.64% accuracy over its base model, BioMistral-7B. However, open-source reasoning-based models encounter difficulties due to token limitations during generation. Manual inspection of the generated text revealed that these models were still in the reasoning process after generating 8,192 tokens, resulting in incomplete answers which negatively impacts their performance.

Multiple-choice question with multiple answers. Table 4 shows the performance of all evaluated LLMs on the MCQM subset of the MediQAI dataset. The best results are achieved by o3 with 55.05 EMR and 79.7 Hamming, followed closely by DeepSeek-R1 (48.88/77.54). Vanilla models such as DeepSeek-V3 and GPT-4o trail by 3–5 EMR points, indicating that additional reasoning supervision yields substantial gains. Distilled checkpoints of DeepSeek-R1 show significant performance drops (e.g. –28 EMR for DeepSeek-R1-Distill-Llama-70B, and –46 EMR for DeepSeek-R1-Distill-Llama-8B), highlighting the trade-off imposed by aggressive model

Model	ROUGE-1			BLEU-4			BERTScore			LLM-as-Judge		
	U	R	Avg.	U	R	Avg.	U	R	Avg.	U	R	Avg.
Baseline	8.21	7.77	7.93	0.89	0.51	0.65	72.58	72.55	72.56	10.12	10.03	10.06
<i>Reasoning LLMs</i>												
o3	<u>17.61</u>	<u>15.6</u>	<u>16.34</u>	<u>2.56</u>	<u>1.5</u>	<u>1.89</u>	77.65	76.48	76.91	87.4	79.07	82.16
DeepSeek-R1	17.8	15.97	16.65	2.68	1.67	2.04	<u>77.63</u>	<u>76.13</u>	<u>76.69</u>	<u>80.26</u>	<u>70.78</u>	<u>74.29</u>
DeepSeek-R1-Distill-Llama-70B	14.41	12.57	13.26	1.85	1.43	1.58	68.64	65.3	66.54	73.16	61.3	65.7
HuatuoGPT-o1-8B	8.23	7.94	8.05	0.64	0.45	0.52	67.56	62.98	64.68	45.32	36.24	39.61
FineMedLM-o1	9.55	9.72	9.66	0.94	0.64	0.75	70.88	70.61	70.71	29.05	24.33	26.08
DeepSeek-R1-Distill-Llama-8B	5.76	5.41	5.54	0.56	0.35	0.43	60.17	55.75	57.39	21.24	16.12	18.02
DeepSeek-R1-Distill-Qwen2.5-7B	4.75	5.23	5.05	0.45	0.34	0.38	52.0	46.55	48.57	11.7	9.27	10.17
<i>Vanilla LLMs</i>												
GPT-4o	16.29	14.53	15.18	2.41	1.29	1.71	76.47	75.55	75.89	77.43	63.77	68.83
DeepSeek-V3	15.24	15.39	15.33	2.06	1.36	1.62	74.95	74.86	74.89	60.37	47.37	52.19
Qwen2.5-72B-Instruct	14.65	13.26	13.77	2.14	1.11	1.49	75.33	74.2	74.62	66.87	55.15	59.49
Llama3.3-70B-Instruct	14.53	13.59	13.94	1.7	1.02	1.27	74.56	73.49	73.89	53.32	43.77	47.31
BioMistral-7B	6.53	8.61	7.84	0.67	0.61	0.63	44.34	53.95	50.39	13.64	13.23	13.38
Llama-3.1-8B-UltraMedical	4.04	3.91	3.96	0.44	0.26	0.33	69.28	67.03	67.87	27.7	20.96	23.46
BioMistral-7B-SFT	5.69	5.59	5.63	0.6	0.38	0.47	73.75	73.53	73.61	23.86	24.38	24.19

Table 5. Performance of LLMs on the MediQAI-OEQ subset. The scores, obtained in zero-shot, are measured with ROUGE-1, BLEU-4, BERTScore and LLM-as-Judge.

compression. In this task, open-source reasoning-based models also face the issue of still being in the reasoning phase after generating 8,192 tokens, which negatively impacts their performance and contributes to their difficulty in exceeding the random baseline score.

Open-ended question with short-answer. Table 5 shows the performance of all evaluated LLMs on the OEQ subset of the MediQAI dataset. For free-text answers, the performance gap widens: o3 achieves 82.16 on the LLM-as-Judge metric, versus 74.29 for DeepSeek-R1 and 68.83 for GPT-4o. Overlap metrics (ROUGE, BLEU and BERTScore) tend to compress differences and often yield trends that diverge from those observed with the LLM-as-Judge metric. For example, DeepSeek-R1 outperforms o3 on ROUGE and BLEU scores, while the opposite is observed with the LLM-as-Judge metric. We also observed that distilled reasoning models from the DeepSeek series often reformulate the question as a multiple-choice question (MCQ) during their reasoning process, creating candidate options. This behavior poses a challenge when parsing the generated text to extract the final answer. Instead of providing a free-text response, the model tends to return the letter of one of the candidate options it created during reasoning, without necessarily including the corresponding text. This phenomenon may partly explain the comparable performance of reasoning models to vanilla models, despite their reasoning capabilities.

Medical reasoning performance. Across all QA tasks, we observe a consistent performance gap between questions that require multi-step reasoning and those assessing factual recall or medical understanding. Averaged over all model-task combinations reported in Tables 3–5, accuracy on reasoning question is 5.12 points lower than understanding questions. The performance gap varies across tasks: it is largest on OEQ (7.54 points), and similar for MCQU (3.90) and MCQM (3.93). Reasoning-based models mitigate this gap to some extent but do not eliminate it. On MCQU and MCQM, the average performance gap for reasoning-augmented models is 2.15 and 2.02 points respectively, compared to 5.49 and 5.55 for vanilla models. In contrast, the OEQ task shows a large gap for both model types: 7.79 for vanilla and 7.29 for reasoning models. To illustrate, on the OEQ task, GPT-4o show a performance gap of 13.66 points between understanding and reasoning questions, which is reduced to 8.33 with its reasoning-enhanced variant, o3. A similar trend is observed for the DeepSeek family: DeepSeek-V3 shows a gap of 13.0, whereas DeepSeek-R1 narrows this to 9.48.

Figure 11 presents paired performance comparisons across three model families on both *Understanding* and *Reasoning* questions:

- o3 vs. GPT-4o
- DeepSeek-R1 vs. DeepSeek-V3
- DeepSeek-R1-Distill-Llama-70B vs. Llama-3.3-70B-Instruct

Two consistent trends emerge across comparisons: (i) every model performs better on *Understanding* than on *Reasoning* questions, except for DeepSeek-R1-Distill-Llama-70B on MCQU and MCQM subsets, and (ii) when comparing each reasoning model to its base version, the performance improvement is larger on reasoning questions than on understanding questions. These differences underscore the impact of inference-time reasoning techniques. On MCQU, the average performance gain for reasoning question across the three model families is 10.67, compared to 4.02 for understanding. Similar trends are observed in MCQM, with gains of

LLM-as-Judge Prompt

[System]

Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the French medical question displayed below. Your evaluation should consider clinical correctness, factual coverage and the impact of differences between the answers on patient safety and care. You will be given a reference answer (Expert-provided answer) and the assistant's answer (Model-generated answer). Your job is to evaluate how closely a Model-generated answer aligns with an Expert-provided answer. Base your judgment only on the Expert's provided answer, and never rely on your own medical knowledge or external resources. Begin your evaluation by comparing the assistant's answer with the reference answer. Avoid any position biases and ensure that the order in which the responses were presented does not influence your decision. Do not allow the length of the responses to influence your evaluation. Be as objective as possible. After providing your explanation, you must rate the response on a scale of 1 to 10 by strictly following this format "[[rating]]", for example: "Rating: [[5]]".

[Medical Question]

{question}

[The Start of Expert Answer]

{answer_ref}

[The End of Expert Answer]

[The Start of Assistant Answer]

{answer_a}

[The End of Assistant Answer]

Fig. 10 Prompt for LLM-as-judge evaluation of LLMs on the OEQ subset.

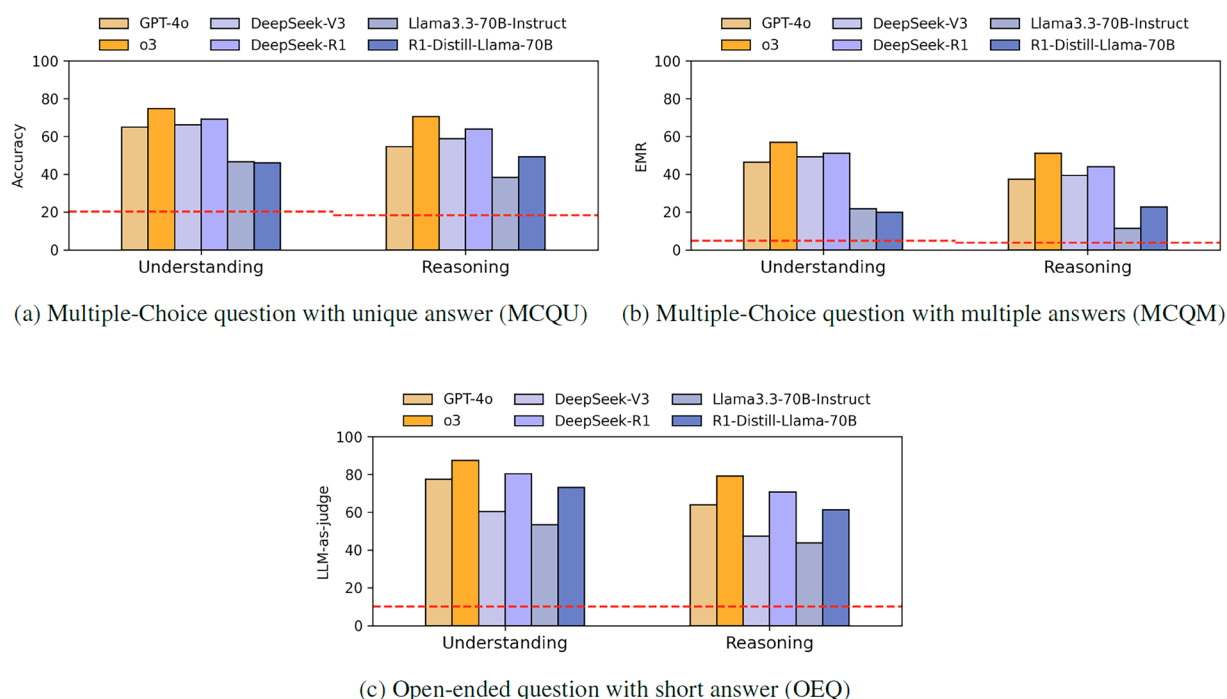


Fig. 11 Performance of three groups of models (OpenAI, DeepSeek and Llama) on all subsets of MediQA. The horizontal line corresponds to the random baseline.

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuogPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Owen2.5-7B	GPT-4o	DeepSeek-V3	Owen2.5-72B-Instruct	Llama3.3-70B-Instruct	Llama-3-70B-UltraMedical	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Anatomy	77.78	88.89	33.33	33.33	0.0	16.67	33.33	77.78	83.33	50.0	38.89	50.0	22.22	11.11	22.22
Biochemistry	71.84	73.79	46.6	27.18	2.91	14.56	14.56	65.05	65.05	52.43	43.69	33.01	14.56	12.62	30.1
Cardiology	78.59	74.31	52.6	26.61	2.45	6.42	11.01	66.67	67.89	46.48	46.48	37.61	12.23	12.84	27.83
Cytology	16.67	16.67	16.67	0.0	0.0	0.0	50.0	16.67	16.67	16.67	16.67	16.67	16.67	0.0	16.67
Dermatology	83.2	73.6	50.4	16.8	4.8	5.6	14.4	70.4	76.0	51.2	52.0	38.4	14.4	13.6	41.6
Embryology	66.67	66.67	33.33	33.33	0.0	0.0	33.33	66.67	66.67	66.67	66.67	0.0	33.33	33.33	66.67
Endocrinology and Metabolism	72.95	67.15	48.31	21.74	1.93	7.25	8.21	58.94	61.84	42.51	41.06	38.65	11.59	13.53	25.6
Epidemiology	53.61	49.48	39.18	20.62	4.12	12.37	22.68	43.3	45.36	40.21	34.02	25.77	10.31	8.25	19.59
Forensic Medicine and Toxicology	72.73	47.47	43.43	20.2	3.03	10.1	15.15	53.54	51.52	36.36	35.35	27.27	11.11	11.11	21.21
Genetics	88.89	88.89	55.56	25.0	11.11	12.5	12.5	73.61	81.94	62.5	61.11	48.61	12.5	25.0	31.94
Gynecology and Obstetrics	68.93	62.86	45.0	14.29	1.79	5.0	8.93	52.14	55.0	44.29	41.79	40.36	10.36	12.14	28.57
Hematology	80.76	71.61	52.05	21.77	4.73	6.31	9.78	62.78	67.19	47.95	42.59	38.17	16.72	13.88	33.12
Hepato-Gastroenterology	64.03	59.71	41.01	23.38	4.32	6.12	12.23	52.88	56.47	41.01	40.29	34.17	12.23	11.51	24.1
Histology	76.06	60.56	43.66	16.9	4.23	14.08	15.49	63.38	59.15	46.48	45.07	30.99	8.45	16.9	18.31
Immunology	68.13	70.33	52.75	24.18	3.3	8.79	14.29	62.64	61.54	57.14	43.96	39.56	7.69	13.19	25.27
Infectious Diseases	73.96	66.15	47.92	22.4	2.6	8.33	10.94	62.5	64.58	49.48	40.62	37.5	13.02	7.81	25.52
Microbiology	68.25	63.49	36.51	33.33	3.17	19.05	15.87	66.67	58.73	46.03	47.62	42.86	12.7	12.7	26.98
Neurology	80.79	72.88	53.11	25.99	3.95	8.47	14.12	68.93	75.14	46.89	44.07	40.11	12.43	13.56	19.77
Nephro-Urology	73.57	71.37	48.46	30.84	2.64	5.73	9.69	64.32	68.28	48.02	43.61	41.85	11.89	11.89	27.75
Occupational Medicine	77.57	70.09	41.12	20.56	2.8	10.28	13.08	55.14	62.62	38.32	42.99	41.12	9.35	13.08	28.97
Ophthalmology	77.57	70.09	41.12	20.56	2.8	10.28	13.08	55.14	62.62	38.32	42.99	41.12	9.35	13.08	28.97
Orthopedics	70.93	64.53	52.33	25.58	3.49	6.98	15.7	58.72	64.53	47.67	45.93	33.72	12.79	10.47	28.49
Otorhinolaryngology (ENT)	81.89	75.59	55.12	25.98	3.15	8.66	11.81	73.23	72.44	50.39	47.24	44.09	9.45	17.32	37.8
Parasitology	78.26	63.04	43.48	28.26	8.7	6.52	13.04	65.22	63.04	45.65	50.0	52.17	4.35	17.39	23.91
Continued															

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuoGPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Owen2.5-7B	GPT-4o	DeepSeek-V3	Owen2.5-72B-Instruct	Llama3.3-70B-Instruct	Llama-3-70B-UltraMedical	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Pathological Anatomy	66.67	61.54	37.18	25.64	6.41	7.69	14.1	53.85	53.85	47.44	35.9	37.18	14.1	24.36	15.38
Pediatrics	69.77	66.51	47.44	23.26	6.05	9.77	14.88	59.07	60.93	50.23	39.07	33.49	12.56	13.49	30.7
Pharmacology	72.22	66.67	53.7	35.19	7.41	16.67	14.81	62.96	61.11	55.56	57.41	37.04	14.81	12.96	37.04
Physiology	82.61	82.61	50.72	17.39	1.45	15.94	18.84	82.61	84.06	52.17	50.72	49.28	13.04	10.14	28.99
Pulmonology	66.13	58.87	44.76	21.77	4.84	8.06	11.29	54.44	56.85	42.74	42.34	36.69	12.9	12.5	29.44
Psychiatry	59.86	48.59	38.73	21.83	1.41	9.15	12.68	50.0	45.77	28.87	42.96	37.32	11.97	9.86	29.58
Radiology	75.0	71.43	25.0	7.14	0.0	3.57	17.86	64.29	71.43	39.29	28.57	28.57	10.71	10.71	17.86
Rheumatology	74.64	68.42	45.45	28.23	4.78	6.22	13.88	57.42	63.64	36.84	37.8	39.23	11.48	13.88	24.88

Table 6. Performance by medical subject on the MediQAL-MCQU subset. The scores, obtained in zero-shot, are measured with Accuracy.

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuoGPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Qwen2.5-7B	GPT-4o	DeepSeek-V3	Qwen2.5-72B-Instruct	Llama3.3-70B-Instruct	Llama-3-70B-UltraMedical	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Biochemistry	46.67	48.89	22.22	11.11	0.0	0.0	2.22	46.67	51.11	42.22	20.0	22.22	0.0	0.0	0.0
Cardiology	62.03	54.43	27.85	8.44	1.27	2.11	2.95	49.79	48.52	29.96	18.57	23.21	0.84	5.49	5.41
Dermatology	66.67	58.49	27.04	4.4	0.0	2.52	4.4	47.8	54.09	35.22	17.61	16.35	1.26	3.77	3.28
Endocrinology and Metabolism	48.89	45.0	18.89	8.89	1.67	1.67	2.22	39.44	45.56	29.44	19.44	21.11	1.67	8.89	6.66
Epidemiology	46.15	38.46	21.54	10.77	1.54	1.54	10.77	35.38	44.62	23.08	15.38	27.69	1.54	9.23	2.41
Forensic Medicine and Toxicology	47.62	27.62	19.05	2.86	1.9	2.86	1.9	35.24	32.38	24.76	18.1	13.33	0.0	2.86	1.75
Genetics	71.64	67.16	29.85	13.43	1.49	2.99	4.48	61.19	65.67	41.79	32.84	29.85	1.49	7.46	5.65
Gynecology and Obstetrics	50.27	37.3	16.22	7.57	0.54	4.86	2.16	36.22	37.3	25.95	23.24	14.59	2.7	3.78	3.37
Hematology	62.96	58.33	25.93	8.8	0.0	2.31	2.31	48.15	50.93	31.94	14.35	17.59	0.46	6.48	4.35
Hepato-Gastroenterology	50.0	45.65	16.96	6.09	1.3	1.3	3.48	38.26	42.61	25.65	16.09	15.22	0.0	1.74	1.29
Immunology	55.22	53.73	23.88	10.45	1.49	2.99	1.49	46.27	50.75	35.82	34.33	29.85	0.0	5.97	2.89
Infectious Diseases	47.14	41.43	16.43	5.0	0.71	5.0	0.71	35.71	42.14	24.29	17.86	15.0	1.43	4.29	1.95
Microbiology	65.79	60.53	36.84	15.79	0.0	0.0	0.0	60.53	71.05	31.58	36.84	21.05	2.63	5.26	4.99
Nephro-Urology	56.34	52.58	23.0	4.69	1.88	0.94	1.88	43.19	45.54	31.92	19.25	17.37	0.0	5.16	4.55
Neurology	63.29	53.16	22.78	10.13	1.27	1.9	1.9	55.06	52.53	37.97	20.89	28.48	0.63	6.33	4.95
Occupational Medicine	38.33	28.33	8.33	6.67	0.0	1.67	5.0	30.0	31.67	18.33	6.67	10.0	0.0	1.67	0.51
Ophthalmology	59.23	57.69	27.69	6.92	1.54	0.77	0.0	47.69	54.62	36.92	26.15	30.0	2.31	1.54	2.12
Orthopedics	54.37	41.75	22.33	3.88	0.97	1.94	2.91	40.78	43.69	31.07	11.65	13.59	0.97	4.85	3.11
Otorhinolaryngology (ENT)	62.77	54.01	14.6	8.76	1.46	4.38	1.46	51.82	46.72	30.66	19.71	19.71	2.92	5.11	4.88
Parasitology	56.0	64.0	24.0	16.0	0.0	0.0	0.0	56.0	56.0	44.0	28.0	32.0	0.0	4.0	3.14
Pathological Anatomy	43.4	41.51	9.43	7.55	1.89	1.89	1.89	28.3	32.08	20.75	16.98	15.09	0.0	7.55	1.38
Pediatrics	52.41	47.59	17.93	5.52	0.69	2.76	1.38	42.07	42.07	27.59	11.03	16.55	1.38	4.14	3.88
Continued															

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuoGPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Qwen2.5-7B	GPT-4o	DeepSeek-V3	Qwen2.5-72B-Instruct	Llama3.3-70B-Instruct	Llama-3-70B-UltraMedical	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Pharmacology	49.23	46.15	23.08	13.85	3.08	6.15	3.08	47.69	50.77	27.69	20.0	20.0	0.0	6.15	5.64
Pulmonology	45.97	43.13	14.69	4.27	0.95	0.95	0.95	34.12	39.34	27.49	14.22	15.64	1.42	6.64	6.33
Psychiatry	47.88	42.42	16.36	6.67	0.0	1.82	1.21	36.97	36.97	26.06	9.09	19.39	0.61	4.85	4.75
Rehabilitation	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0
Rheumatology	64.48	60.11	21.31	7.65	1.64	1.64	1.64	55.19	56.28	34.43	24.04	21.86	0.0	4.37	2.05

Table 7. Performance by medical subject on the MediQAL-MCQM subset. The scores, obtained in zero-shot, are measured with EMR.

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuogPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Qwen2.5-7B	GPT-4o	DeepSeek-V3	Qwen2.5-72B-Instruct	Llama3.3-70B-Instruct	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Bacteriology	98.82	95.29	85.88	60.0	51.18	27.65	21.76	87.06	75.88	86.47	51.76	5.29	45.88	41.18
Cardiology	80.74	73.46	65.35	39.91	23.23	21.29	11.2	69.95	51.71	59.63	47.93	9.91	22.07	27.28
Dermatology	80.74	71.48	62.22	32.96	19.63	18.15	7.22	62.78	47.04	55.0	40.93	12.41	20.0	28.52
Emergency Medicine	74.13	65.05	59.82	33.49	22.94	16.88	8.35	58.53	42.48	54.04	43.49	11.83	20.92	21.93
Endocrinology and Metabolism	66.25	60.42	45.0	28.75	22.5	10.42	10.0	56.25	44.58	50.0	33.75	7.08	13.75	15.83
Forensic Medicine and Toxicology	77.79	66.63	59.53	27.44	21.28	17.56	9.53	59.19	46.28	49.77	43.37	13.49	17.21	18.02
Genetics	85.54	80.92	74.15	44.77	30.46	19.54	20.15	74.92	56.62	66.15	50.92	5.38	26.92	19.38
Gynecology and Obstetrics	81.01	70.2	64.9	43.96	23.36	20.27	12.55	68.72	48.46	62.75	51.95	16.85	26.44	26.64
Hematology	86.42	78.54	67.59	39.42	20.15	17.59	9.27	69.85	60.22	59.12	49.56	12.7	22.34	24.96
Hepato-Gastroenterology	74.71	66.32	52.94	33.82	25.15	17.21	7.21	55.88	42.06	50.15	41.03	10.0	20.29	24.85
Immunology	76.88	78.12	75.0	56.88	50.62	26.25	14.38	70.0	59.38	64.38	54.38	22.5	38.75	37.5
Infectious Diseases	73.75	70.68	57.39	37.5	22.39	14.89	8.86	62.73	44.32	53.18	40.57	11.36	17.39	22.39
Intensive Care	76.95	72.79	66.1	35.0	23.31	15.91	7.99	64.87	45.78	58.31	44.48	15.45	21.75	24.87
Nephro-Urology	83.58	75.85	66.11	42.02	23.26	18.13	10.62	70.1	54.09	59.84	48.39	11.14	24.77	26.84
Neurology	79.2	73.13	64.2	36.53	20.47	18.33	9.0	63.73	49.0	56.87	45.2	13.47	20.67	25.13
Occupational Medicine	54.38	50.0	38.75	22.5	16.88	11.25	9.38	43.75	32.5	30.0	30.62	10.62	8.75	19.38
Ophthalmology	77.1	69.03	56.13	41.94	23.23	19.68	6.45	60.32	46.13	62.26	40.0	11.29	21.94	25.81
Orthopedics	77.31	68.28	52.8	33.23	21.83	17.85	10.22	61.18	45.7	51.08	42.47	11.51	23.66	23.12
Otorhinolaryngology (ENT)	80.95	73.73	67.54	44.21	26.83	18.33	9.52	66.59	54.05	61.03	48.89	18.41	22.78	31.83
Parasitology	94.35	93.04	86.09	54.78	42.17	29.57	17.39	86.09	73.48	84.78	53.04	7.39	38.7	33.91
Pediatrics	85.11	74.47	53.62	34.04	23.4	18.3	9.15	65.74	43.83	52.55	45.74	9.57	18.94	23.4
Pediatric Cardiology	81.08	71.62	62.76	35.93	23.88	15.22	11.63	67.07	54.18	54.29	46.57	9.49	22.02	19.62
Pharmacology	68.89	75.56	62.22	35.56	18.89	32.22	8.89	62.22	44.44	48.89	26.67	10.0	11.11	11.11
Pharmacy	86.51	79.29	71.96	43.67	30.98	19.36	9.07	75.04	54.4	65.79	49.68	17.03	26.56	25.51
Psychiatry	78.5	69.12	62.38	36.88	21.25	22.0	9.12	63.62	46.0	60.0	47.12	14.62	20.25	30.12
Public Health	71.05	64.74	43.68	36.32	32.63	27.37	18.42	50.53	39.47	52.11	40.0	18.42	22.63	21.58
Pulmonology	80.6	72.77	61.2	43.59	25.43	18.26	9.08	65.0	48.53	59.84	47.61	17.93	19.67	30.0
Continued														

Medical Subject	o3	DeepSeek-R1	DeepSeek-R1-Distill-Llama-70B	HuatuoGPT-o1-8B	FineMedLM-o1	DeepSeek-R1-Distill-Llama-8B	DeepSeek-R1-Distill-Qwen2.5-7B	GPT-4o	DeepSeek-V3	Qwen2.5-72B-Instruct	Llama3.3-70B-Instruct	BioMistral-7B	Llama-3.1-8B-UltraMedical	BioMistral-7B-SFT
Rehabilitation	75.0	61.67	53.33	45.0	35.0	28.33	10.0	50.0	33.33	66.67	50.0	10.0	23.33	18.33
Rheumatology	75.14	65.14	58.38	35.41	25.41	16.22	8.11	58.92	46.22	48.92	37.03	9.73	17.3	24.05
Semiology	95.0	95.0	85.0	60.0	15.0	5.0	10.0	85.0	65.0	65.0	50.0	0.0	60.0	10.0

Table 8. Performance by medical subject on the MediQAL-OEQ subset. The scores, obtained in zero-shot, are measured with LLM-as-judge.

9.87 on reasoning versus 3.51 on understanding. For the OEQ task, performance improvements are substantial in both categories with 16.57 for understanding and 18.75 for reasoning questions. These findings suggest that inference-time techniques, even without access to domain-specific adaptation, can significantly enhance complex medical reasoning. Nonetheless, even state-of-the-art LLMs remain well below human-level clinical reasoning in zero-shot settings. For downstream applications in healthcare, these models will require external verification or human oversight.

Medical subjects performance. To better understand the strengths and weaknesses of LLMs on our dataset, we analyzed their performance across individual medical subjects for each QA task (see Tables 6–8). In the MCQU task, the models performed best on subjects such as genetics, anatomy, dermatology, physiology, otorhinolaryngology (ENT), ophthalmology, neurology and hematology, all achieving over 80% accuracy. Conversely, subjects like cytology (notably low at 16.67% due to limited examples in the dataset), epidemiology, and psychiatry showed the lowest performance with accuracy below 60%. In the MQCM task, the easiest subjects for LLMs were dermatology, genetics, and microbiology (all above 65% EMR), while rehabilitation, occupational medicine, and pathological anatomy were the most challenging, with scores under 40% EMR. Finally, in the OEQ task, the best-performing subjects were bacteriology, parasitology, and semiology, each with LLM-as-Judge scores above 90%, whereas occupational medicine and endocrinology and metabolism were among the lowest, with score falling below 70%. These results highlight that LLMs' capabilities vary significantly by medical domain and question type, with certain specialized or interdisciplinary fields remaining particularly challenging.

To further investigate these subject-level divergences, we examined the relationship between performance and several dataset characteristics, including the number of questions per subject, average question length, and the proportion of reasoning questions. Across all tasks, performance showed little association with the number of questions per subject. However, task-dependent patterns emerged for other dataset characteristics. For MCQU, correlations between performance and both question length and reasoning proportion were negligible. In contrast, for MCQM, moderate negative correlations were observed between performance and both average question length ($\rho = -0.28$) and the proportion of reasoning questions ($\rho = -0.38$), suggesting that increased reasoning proportion contribute to subject difficulty and impact performance. This effect was most pronounced in the OEQ task, where performance across models was strongly negatively correlated with the proportion of reasoning questions ($\rho = -0.70$) and average question length ($\rho = -0.46$). Consistent with these findings, agreement in subject-wise performance rankings across models increased with task complexity, from moderate agreement in MCQU (Kendall's $W = 0.31$), to slightly higher agreement in MCQM ($W = 0.36$), and strong agreement in OEQ ($W = 0.51$).

These observations suggest that subject-level performance differences arise from an interaction between subject characteristics and task-specific reasoning demands. Residual differences may also be influenced by factors not directly captured in the dataset or evaluation setup, such as variation in model architectures or differences in the composition of pre-training and fine-tuning data across models.

Environmental impact of models. This study required substantial computational resources, for a total of approximately 4,000 hours on A100 80GB GPUs for open source models (DeepSeek-R1-Distill, Llama3, Qwen2.5, BioMistral, FineMedLM, and HuatuoGPT). These resources were dedicated to model evaluation, experimentation with various models, and debugging. According to documentation from the Jean Zay supercomputer, the total environmental cost amounted to 1,036,000 Wh or 59,05 kgCO₂eq. Additionally, the total inference cost on API for data augmentation strategies, the LLM-as-Judge evaluation, and the zero-shot evaluation of o3, GPT-4o, DeepSeek-R1, and DeepSeek-V3 on the three tasks amounted to a total environmental cost of 43.36 kgCO₂eq, according to Ecologits³⁹.

Data availability

The dataset is publicly available in JSON format on Zenodo (<https://doi.org/10.5281/zenodo.18220039>)²⁵ and Hugging Face (<https://huggingface.co/datasets/ANR-MALADES/MediQAI>).

Code availability

All python scripts to reproduce the evaluation of models on MediQAI are available on Github (<https://github.com/abazoge/MediQAI>).

Received: 28 July 2025; Accepted: 21 January 2026;

Published online: 05 February 2026

References

- Lawrence, K. Q. *et al.* Large language model benchmarks in medical tasks. Preprint, arXiv:2410.21348 (2024).
- Hendrycks, D. *et al.* Measuring massive multitask language understanding. In *International Conference on Learning Representations*. (2021).
- Wang, Y. *et al.* MMLU-pro: A more robust and challenging multi-task language understanding benchmark. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. (2024).
- Shi, X. *et al.* Medical dialogue system: A survey of categories, methods, evaluation and challenges. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2840–2861, Bangkok, Thailand. Association for Computational Linguistics. (2024).
- Vilares, D. & Gómez-Rodríguez, C. HEAD-QA: A healthcare dataset for complex reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 960–966, Florence, Italy. Association for Computational Linguistics. (2019).
- Jin, D. *et al.* What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, **11**(14) (2021).
- Pal, A., Kumar Umapathi, L. & Sankarasubbu, M. Medmqca: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Proceedings of the Conference on Health, Inference, and Learning*, volume 174 of *Proceedings of Machine Learning Research*, 248–260. PMLR. (2022).

8. Jin, Q., Dhingra, B., Liu, Z., Cohen, W. & Lu, X. PubMedQA: A Dataset for Biomedical Research Question Answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2567–2577, Hong Kong, China. Association for Computational Linguistics. (2019).
9. Li, J., Zhong, S. & Chen, K. 2021. MLEC-QA: A Chinese Multi-Choice Biomedical Question Answering Dataset. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 8862–8874, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics. (2021).
10. Michael Bean, A., Korgul, K., Krones, F., McCraith, R. & Mahdi, A. Do large language models have shared weaknesses in medical question answering? In *Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond*. (2024).
11. Alonso, I., Oronoz, M. & Agerri, R. Medexpqa: Multilingual benchmarking of large language models for medical question answering. *Artificial Intelligence in Medicine* **155**, 102938 (2024).
12. Labrak, Y. *et al.* FrenchMedMCQA: A French multiple-choice question answering dataset for medical domain. In *Proceedings of the 13th International Workshop on Health Text Mining and Information Analysis (LOUHI)*, 41–46, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics. (2022).
13. Kaddari, Z. & Bouchentouf, T. FrBMedQA: the first French biomedical question answering dataset. *IAES International Journal of Artificial Intelligence (IJ-AI)* **11**(4), 1588–1595 (2022).
14. Nachane, S. S. *et al.* Few shot chain-of-thought driven reasoning to prompt llms for open ended medical question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 542–573, Miami, Florida, USA. Association for Computational Linguistics. (2024).
15. Quercia, A., Zaghir, J., Lovis, C. & Gaudet-Blavignac, C. MedFrenchmark, a small set for benchmarking generative LLMs in medical french. *Stud. Health Technol. Inform.* **316**, 601–605 (2024).
16. Zuo, Y. *et al.* Medxpertqa: Benchmarking expert-level medical reasoning and understanding. In *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267:80961–80990 (2025).
17. Jin, Y. *et al.* Better to ask in english: Cross-lingual evaluation of large language models for healthcare queries. In *Proceedings of the ACM Web Conference 2024*, WWW'24, 2627–2638, New York, NY, USA. Association for Computing Machinery (2024).
18. Dey, K., Tarannum, P., Hasan, M. A., Razzak, I. & Naseem, U. Better to ask in english: Evaluation of large language models on english, low-resource and cross-lingual settings. Preprint, arXiv:2410.13153 (2024).
19. Grattafiori, A. *et al.* The llama 3 herd of models. Preprint, arXiv:2407.21783 (2024).
20. Team, Q. Qwen2.5: A party of foundation models (2024).
21. Jiang, A. Q. *et al.* Mistral 7b. Preprint, arXiv:2310.06825 (2023).
22. Sparck Jones, K. A statistical interpretation of term specificity and its application in retrieval. *Journal of documentation* **28**(1), 11–21 (1972).
23. Labrak, Y. *et al.* DrBERT: A robust pre-trained model in French for biomedical and clinical domains. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 16207–16221, Toronto, Canada. Association for Computational Linguistics (2023).
24. OpenAI. Gpt-4o system card. Preprint, arXiv:2410.21276 (2024).
25. Bazoge, A. MediQAL: A French Medical Question Answering Dataset for Knowledge and Reasoning Evaluation [Data set]. *Zenodo*. <https://doi.org/10.5281/zenodo.18220039> (2025).
26. DeepSeek-AI. Deepseek-v3 technical report. Preprint, arXiv:2412.19437 (2024).
27. Zhang, K. *et al.* Ultramedical: Building specialized generalists in biomedicine. *Advances in Neural Information Processing Systems* **37**, 26045–26081 (2024).
28. Labrak, Y. *et al.* BioMistral: A collection of open-source pretrained large language models for medical domains. In *Findings of the Association for Computational Linguistics: ACL 2024*, 5848–5864, Bangkok, Thailand. Association for Computational Linguistics (2024).
29. OpenAI. Openai o1 system card. Preprint, arXiv:2412.16720 (2024).
30. Guo, D. *et al.* DeepSeek-R1 DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* **645**, 633–638 (2025).
31. Chen, J. *et al.* Towards Medical Complex Reasoning with LLMs through Medical Verifiable Problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria. Association for Computational Linguistics. (2025).
32. Yu, H. *et al.* FinemedLM-o1: Enhancing medical knowledge reasoning ability of LLM from supervised fine-tuning to test-time training. In *Second Conference on Language Modeling*. (2025).
33. Ben Abacha, A., Yim, W.W., Michalopoulos, G & Lin, T. An investigation of evaluation methods in automatic medical note generation. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2575–2588, Toronto, Canada. Association for Computational Linguistics (2023).
34. Lin, C.-Y. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, 74–81, Barcelona, Spain. Association for Computational Linguistics (2004).
35. Papineni, K., Roukos, S., Ward, T & Zhu, W.-J. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics, ACL '02*, 311–318, USA. Association for Computational Linguistics (2002).
36. Zhang*, T., Kishore*, V., Wu*, F., Weinberger, K. Q. & Artzi, Y. Bertscore: Evaluating text generation with bert. In *International Conference on Learning Representations* (2020).
37. Gu, J. *et al.* A survey on llm-as-a-judge. *The Innovation* (2026).
38. Google DeepMind. Introducing Gemini 2.0: our new ai model for the agentic era. <https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/> (2024).
39. Rincé, S. & Banse, A. Ecologits: Evaluating the Environmental impacts of Generative AI. *Journal of Open Source Software* **10**(111), 7471 (2025).

Acknowledgements

This work was performed using HPC resources from GENCI-IDRIS (Grant 2024-AD011013715R2). This work was financially supported by ANR MALADES (ANR-23-IAS1-0005). The authors would also like to thank Pacôme Constant dit Beaufile for his support in evaluating the dataset quality.

Author contributions

All the work presented in this paper was carried out by A.B.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to A.B.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026